

Respiratory Rate: The Forgotten Vital Sign

Respiratory rate (RR) is the number of breaths a person takes per minute, measured by counting chest rises over 60 seconds while the patient is at rest and unaware they are being observed, as conscious observation often alters the breathing pattern. The normal respiratory rate in a healthy adult is **12–20 breaths per minute**. In children, the range is higher and age-dependent, with neonates breathing at 30–60 breaths per minute, while a 5-year-old may breathe at 20–30. Bradypnoea refers to a rate below 12, while tachypnoea describes a rate above 20 in adults. Respiratory rate is arguably the most sensitive early warning sign of clinical deterioration, yet it remains the most frequently underdocumented vital sign in practice. In triage, an RR above 25 breaths per minute is a red flag demanding immediate attention, often indicating sepsis, pulmonary embolism, pneumonia, metabolic acidosis, or impending respiratory failure. An RR below 8 is equally alarming, suggesting opioid toxicity, neurological compromise, or severe metabolic derangement. Most early warning scoring systems, such as the **NEWS2**, assign the highest point weighting to respiratory rate precisely because of its predictive power. Unlike blood pressure or SpO₂, respiratory rate requires no equipment, just a watch and an attentive eye, yet studies consistently show it is charted less accurately than any other vital sign, often estimated rather than counted. In a deteriorating patient, a rising respiratory rate frequently precedes haemodynamic instability by hours, making it a critical window for early intervention.

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References

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